Results ML Mini Project

Sridevi Mudiyanur Ramamurthy

October 2025

Discuss performance of different models. Specifically:

Task 5.1: How does the generalization performance of each model change as a function of the amount of training data and the choice of hyper-parameters?

Across all the experiments, both MLE and MAP models improved as the training data increased, which showed better generalization with larger datasets. But with smaller or imbalanced data samples, MLE tended to overfit, achieving high training but lower validation likelihoods. On the other hand, MAP performed more consistently due to the priors (α, β), which regularized learning and prevented overconfidence.

Varying the pixel prior (β) showed that moderate values (around 10) showed smoother and more stable performance, while very high β (100) caused the model to underfit. The class prior α, showed that moderate α (10–100) offered the best balance, small α led to bias towards frequent classes, while large α led to over-smoothing predictions.

Under imbalanced data (small α_{class}), MAP's validation performance was stronger than MLE, which demonstrated robustness. As balanced data increased (higher α_{class}), both models converged. MAP model with moderate α and β showed the best generalization, especially with limited or imbalanced data.

Task 5.2: Which model would you select for deployment and why?

From all the results, the MAP Naïve Bayes model clearly stands out as the better option for deployment. It performed more consistently across different training sizes, handled data imbalance well and avoided overfitting seen in the MLE model. The use of priors gave the model a reliable learning process, especially when the data was uneven.

MLE model works well when there's plenty of balanced data, but it struggled when the training samples were smaller. Since, real world datasets like EMNIST have imbalances. The MAP model with moderate α (10–100) and β (2–10) provides the best overall performance.